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**Introduction to Computational Chemistry**

**Course Code** - CHY1005

***"PROJECT REPORT"***

**Topic** - Intermolecular Forces and Molecular Interactions

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**Slot** – C11+C12+C13

**Aim:-**

The aim of this project is to understand different types of intermolecular forces and analyze how these forces affect the behaviour of selected molecules such as $H_2O$, $NH_3$, $CO_2$, $CH_4$, and HCl through the development of an interactive educational website.

**Introduction:-**

Intermolecular forces are attractive forces that exist between molecules. These forces are weaker than chemical bonds but play a very important role in determining physical properties such as boiling point, melting point, solubility, viscosity, and molecular structure behaviour.

Understanding intermolecular forces helps explain why some substances are gases while others are liquids or solids at room temperature. In this project, we studied four major types of intermolecular forces and analyzed their presence in selected molecules.

**Intermolecular Forces:-**

Intermolecular forces are forces of attraction or repulsion between neighbouring molecules. They influence many physical properties of substances including boiling point, surface tension, and solubility.

The strength of intermolecular forces varies depending on molecular polarity, size, and structure.

**Main types of intermolecular forces studied in this project:-**

1. London Dispersion Forces
2. Dipole–Dipole Forces
3. Hydrogen Bonding
4. Ion–Dipole Forces

**Types of Intermolecular Forces:-**

**1. London Dispersion Forces**

London dispersion forces are weak temporary attractive forces that occur due to momentary fluctuations in electron distribution around molecules. These forces are present in all molecules but are especially important in nonpolar molecules.

Strength Level: Weakest intermolecular force

Where it occurs: Present in all molecules, especially nonpolar molecules like $CH_4$ and $CO_2$

Example: Methane ($CH_4$)

**2. Dipole–Dipole Forces**

Dipole–dipole forces occur between polar molecules that have permanent dipoles. The positive end of one molecule attracts the negative end of another molecule.

Strength Level: Moderate strength

Where it occurs: Between polar molecules such as HCl

Example: Hydrogen chloride (HCl)

**3. Hydrogen Bonding**

Hydrogen bonding is a strong type of dipole–dipole interaction that occurs when hydrogen is bonded to highly electronegative atoms such as nitrogen, oxygen, or fluorine.

Strength Level: Strong intermolecular force

Where it occurs: In molecules like $H_2O$ and $NH_3$

Example: Water ($H_2O$)

**4. Ion–Dipole Forces**

Ion–dipole forces occur between ions and polar molecules. These forces are important in solutions where ionic compounds dissolve in polar solvents like water.

Strength Level: Strongest among the four discussed forces

Where it occurs: Between ions and polar solvent molecules

Example: Dissolution of salt (NaCl) in water

**Molecule Analysis:-**

1. Water ($H_2O$)

Type of Force: **Hydrogen Bonding**

Reason: Oxygen is highly electronegative and forms strong hydrogen bonds with hydrogen atoms of nearby water molecules.

Polarity: Polar molecule

Boiling Point: High due to strong hydrogen bonding

Application: Used as universal solvent

2. Ammonia ($NH_3$)

Type of Force: **Hydrogen Bonding**

Reason: Nitrogen is electronegative and allows hydrogen bonding between ammonia molecules.

Polarity: Polar molecule

Boiling Point: Higher than expected due to hydrogen bonding

Application: Used in fertilizers and cleaning agents

3. Carbon Dioxide ($CO_2$)

Type of Force: **London Dispersion Forces**

Reason: Linear symmetrical structure makes the molecule nonpolar even though individual bonds are polar.

Polarity: Nonpolar molecule

Boiling Point: Low due to weak dispersion forces

Application: Used in fire extinguishers and carbonated drinks

4. Methane ($CH_4$)

Type of Force: **London Dispersion Forces**

Reason: Symmetrical tetrahedral structure makes methane nonpolar.

Polarity: Nonpolar molecule

Boiling Point: Very low due to weak intermolecular forces

Application: Used as fuel (natural gas)

5. Hydrogen Chloride (HCl)

Type of Force: **Dipole–Dipole Forces**

Reason: Difference in electronegativity between hydrogen and chlorine creates a permanent dipole.

Polarity: Polar molecule

Boiling Point: Moderate due to dipole–dipole interaction

Application: Used in preparation of hydrochloric acid

**Applications of Intermolecular Forces in Daily Life:-**

Intermolecular forces play an important role in many real-life situations such as:

- Water rising in plants through capillary action
- Dissolving salt in water
- Formation of ice structure
- Surface tension of liquids
- Protein folding in biological systems

These forces help explain many physical and biological processes around us.

**Conclusion:-**

This project helped us understand the importance of intermolecular forces and their effect on molecular behavior and physical properties. By analyzing different molecules such as $H_2O$, $NH_3$, $CO_2$, $CH_4$, and HCl, we observed how molecular structure determines the type and strength of intermolecular forces present.

The interactive website developed as part of this project provides a simple and effective way to visualize and learn these concepts.